

Classical Controls

Root Locus

$$t_{r1} \approx 2.2\tau \quad t_{r2} \approx 1.8/\tau \quad t_{s1} \approx 4\tau \quad t_{s2} \approx 4/\zeta\omega_n$$

$$t_P = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$$

$$\%O.S. = M_P \times 100\%$$

$$\frac{Y(s)}{R(s)} = K \frac{1/\tau}{s + 1/\tau}$$

$$M_P = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

$$\zeta = \frac{-\ln(\%O.S./100)}{\sqrt{\pi^2 + \ln^2(\%O.S./100)}}$$

$$\frac{Y(s)}{R(s)} = K \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

1. Initial value: $\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s)$
2. Final value: **iff** all roots of $sF(s)$ are negative real, then $f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$

Robot Control

Nonlinear Feedback Compensator

Typically $\ddot{H}$ omitted due to noisy $\ddot{\theta}$. Gains are typically diagonal matrices. Can be very noisy.

$$\text{OL Dyn: } \ddot{\tau} = \underbrace{I}_{\text{lin}} + \underbrace{H(\ddot{\theta})}_{\text{nonlin.}} \ddot{\theta} + \underbrace{V(\ddot{\theta}, \dot{\theta})}_{\text{nonlin.}} + \underbrace{F(\ddot{\theta})}_{\text{nonlin.}} + \underbrace{B\ddot{\theta}}_{\text{lin}} + \underbrace{G(\ddot{\theta})}_{\text{nonlin}}$$

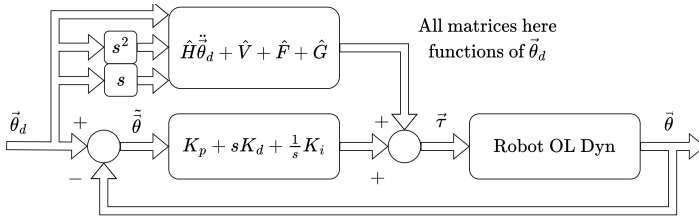
$$\text{Control law: } \ddot{\tau} = \left(K_p + sK_d + \frac{1}{s}K_i\right) \ddot{\theta} + \hat{H}(\ddot{\theta})\ddot{\theta} + \hat{V}(\ddot{\theta}, \dot{\theta}) + \hat{G}(\ddot{\theta}) + \hat{F}(\ddot{\theta})$$

$$\text{CL Dyn: } I\ddot{\theta} + B\ddot{\theta} = \left(K_p + sK_d + \frac{1}{s}K_i\right) \ddot{\theta}$$

Feedforward Computed Torque Control

$$\text{OL Dyn: } \ddot{\tau} = \underbrace{I}_{\text{lin}} + \underbrace{H(\ddot{\theta})}_{\text{nonlin.}} \ddot{\theta} + \underbrace{V(\ddot{\theta}, \dot{\theta})}_{\text{nonlin.}} + \underbrace{F(\ddot{\theta})}_{\text{nonlin.}} + \underbrace{B\ddot{\theta}}_{\text{lin}} + \underbrace{G(\ddot{\theta})}_{\text{nonlin}}$$

$$\text{Control law: } \ddot{\tau} = \left(K_p + sK_d + \frac{1}{s}K_i\right) \ddot{\theta} + \hat{H}(\ddot{\theta}_d)\ddot{\theta}_d + \hat{V}(\ddot{\theta}_d, \dot{\theta}_d) + \hat{F}(\ddot{\theta}_d) + \hat{G}(\ddot{\theta}_d)$$



Inverse Dynamic Control/Feedback Linearization

Centralized control, accelerates $\ddot{\theta}$ to zero. Removes dynamic coupling of error. $\ddot{\theta}_d$ is the desired acceleration, so $\ddot{\theta}_d = \ddot{u}$. With perfect models, $G_p(s) = 1/s^2$, and perfect tracking.

$$\text{OL Dyn: } \ddot{\tau} = H(\ddot{\theta})\ddot{\theta} + \underbrace{V(\ddot{\theta}, \dot{\theta}) + F(\ddot{\theta}) + G(\ddot{\theta})}_{N(\ddot{\theta}, \dot{\theta})}$$

$$\text{Control law: } \ddot{\tau} = \hat{H}(\ddot{\theta})\ddot{u} + \hat{N}(\ddot{\theta}, \dot{\theta})$$

$$\ddot{u} = \left(K_p + sK_d + \frac{1}{s}K_i\right) \ddot{\theta} + \ddot{\theta}_d$$

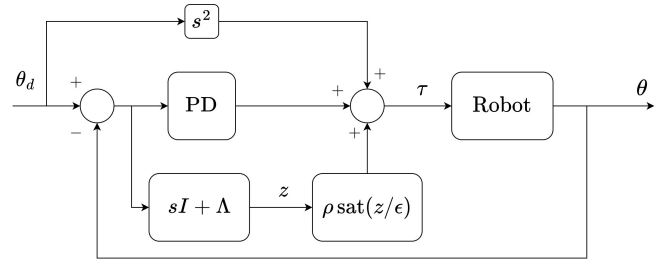
Sliding Mode Control

Pick $\lambda_i \geq 5 \times \text{PD dyn. w/ } \lambda_i$ the time const. of the sliding surface for joint i . Set $\varepsilon > \dot{\theta}$ = width of allowable chatter

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n) \quad \rho \geq \|\text{model error}\|_{\max} = \text{"hardware sat."}$$

$$\text{OL Dyn: } \tau = H(\theta)\ddot{\theta} + N(\theta, \dot{\theta})$$

$$\text{Control law: } \hat{\tau} = \hat{H}(\theta)u + \hat{N}(\theta, \dot{\theta})$$



Adaptive Control

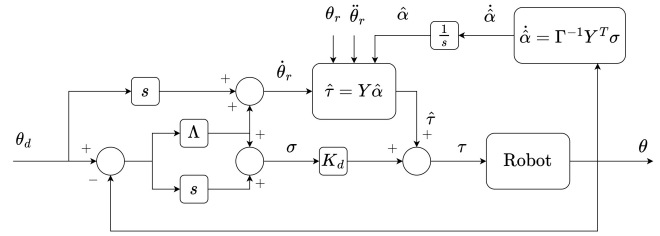
$\hat{Y}$ is chosen based upon the structured uncertainty terms α such that $\hat{\tau} = Y\hat{\alpha}$. Y will then be a function of whatever angles/signals necessary. To guarantee Lyapunov stability, $\ddot{\theta}$ are replaced with $\dot{\theta}$ -reference, any centripetal/coriolis terms get one velocity replaced with dot-reference, and actual signals stay as measured.

$$\tau = \hat{\tau} + K_d(s\dot{\theta} + \Lambda\tilde{\theta}) \quad K_p = K_d\Lambda \quad \sigma = s\dot{\theta} + \Lambda\tilde{\theta}$$

$$\hat{\tau} = Y\hat{\alpha} \quad \dot{\hat{\alpha}} = \Gamma^{-1}Y^T\sigma$$

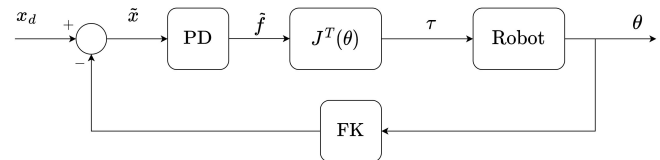
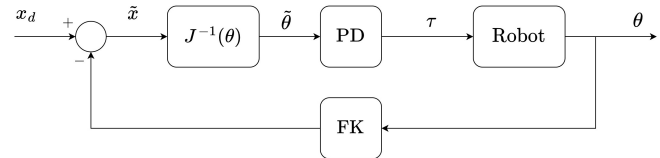
$$\text{OL Dyn: } \tau = H(\theta)\ddot{\theta} + N(\theta, \dot{\theta})$$

$$\text{Control law: } \hat{\tau} = \hat{H}(\theta)u + \hat{N}(\theta, \dot{\theta})$$



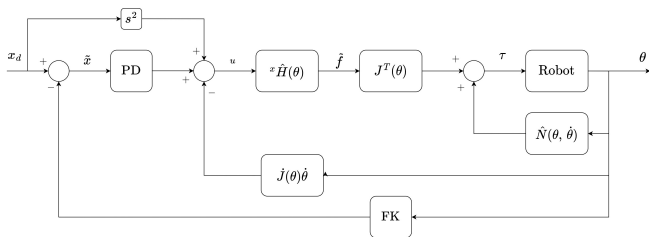
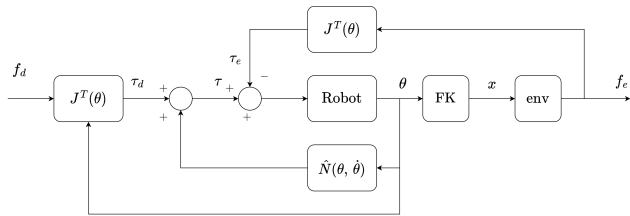
Jacobian Inverse/Transpose Control

Doesn't require IK. $\dot{x} = J(\theta)\dot{\theta} \implies \Delta x \approx J(\theta)\Delta\theta \implies \tilde{\theta} \approx J^{-1}(\theta)\tilde{x}$

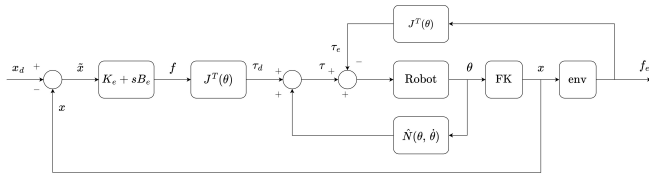


IDC in Op-Space

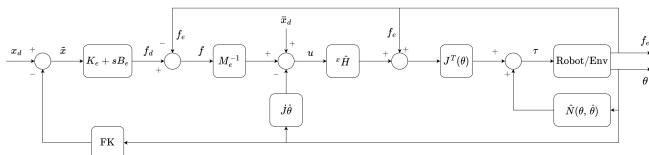
Sys. behaves like $1/s^2$ ${}^x\hat{H}(\theta) = J^{-T}(\theta)\hat{H}(\theta)J^{-1}(\theta)$

Feedforward Force Control (direct)Stiffness/Damping Control (indirect)

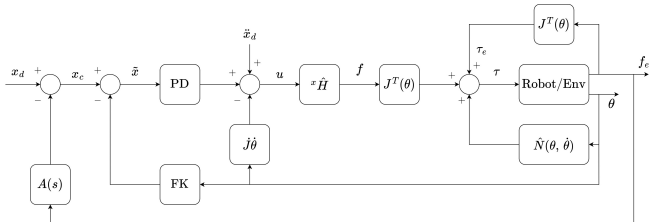
Programmable stiffness/damping. Feels like spring/damper at EE

Impedance Control (indirect)

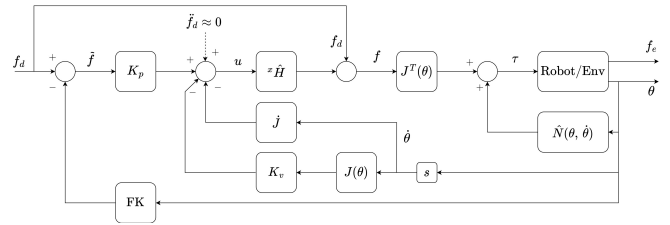
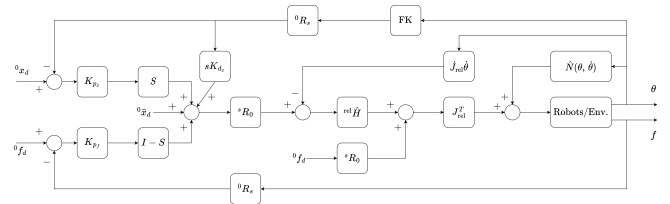
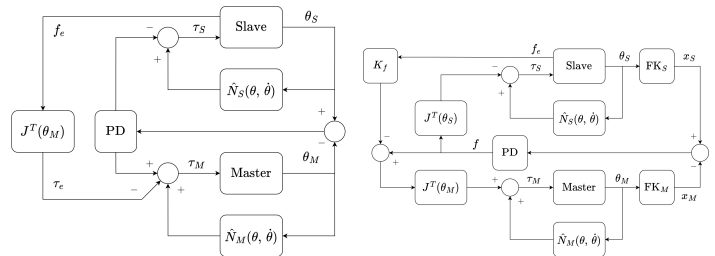
Softening PD gains results in poor tracking and dist. rej. If robot is not back-drivable, works poorly, as hard to displace.

Admittance Control (direct)

PD gains stay high. C altered to affect behavior. NOT Lyapunov stable. Design $A(s)$ to have desired 2nd order characteristics.

Direct Force Control (direct)

NOT Lyapunov stable, as $k_p k_w$ appears in char. eq./loop gain

Hybrid/Multi-Arm Control (direct)Bilateral (Servo Joint-Space [L]. Servo/Direct Force Op-Space [R])Types of Force Control

	Force Feedback	No Force Feedback
Direct	Direct Force Control	FF Force Control
Indirect	Impedance/Admittance	Stiffness Control

Relative Task Space

Jacobians map wrenches ${}^a w = {}^a J_b^T {}^b w$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_{rel} \end{bmatrix} = \begin{bmatrix} J_1 & 0 \\ -J_1 & J_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = J_{rel} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \quad \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = J_{rel}^T \begin{bmatrix} f_{ext} \\ f_{int} \end{bmatrix}$$

Constraint Table

All components of task-space must show up in each col, and row.

	Kinematic	Static
Natural	keep obj. together	frictionless env.
Artificial	pos. control	force control